

MH3100 Real Analysis I

Tutorial 3 Solutions

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Overview

- **Lecture coverage:** Lectures 7–9 — subsequences and the Theorem on Subsequential Limits (Lecture 7), the Divergence Criterion (stated in Lecture 7, completed in Lecture 8), the Bolzano-Weierstrass Theorem (Lecture 8), and the Monotone Convergence Lemma / Monotone Convergence Theorem (Lecture 9).
- **Questions:** 9 questions from Tutorial 3 (Q1–Q5, plus Additional Questions A1–A4); solutions extend the official Tutorial 3 slides and Answers-and-Hints sheet.
- **Notation:** $\mathbb{N} = \{1, 2, 3, \dots\}$ (so $0 \notin \mathbb{N}$). A *subsequence* of $\{a_n\}_{n=1}^{\infty}$ is a sequence $a_{n_1}, a_{n_2}, a_{n_3}, \dots$, where $n_1 < n_2 < n_3 < \dots$ is a strictly increasing sequence of natural numbers (Lecture 7, deck p. 6); it is *proper* if $n_k \neq k$ for some $k \in \mathbb{N}$. We use this course definition of “proper” throughout, in particular in Question 1(a).
- **Key facts used repeatedly:** the Divergence Criterion is stated for a *bounded* sequence $\{a_n\}$: it is divergent if and only if it has two subsequences converging to different limits (Lecture 7, deck p. 14). The Monotone Convergence Theorem is proved via Bolzano-Weierstrass together with the Monotone Convergence Lemma (Lecture 9). Every sequence is, by definition, either convergent or divergent, never both – this dichotomy is used repeatedly to turn “not divergent” into “convergent.”

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Question 1 (True/False – Subsequences and Convergence)

Problem

Determine if each of the following statements is true.

- (a) If every proper subsequence of a sequence $\{a_n\}$ converges, then $\{a_n\}$ converges as well.
- (b) If every convergent subsequence of a sequence $\{a_n\}$ converges to a , then $\{a_n\}$ converges to a as well.
- (c) If a sequence $\{a_n\}$ contains a divergent subsequence, then $\{a_n\}$ diverges.

(Tutorial 3, Question 1; Lectures 7–9. Part (a) is also Sample Midterm Question 1(c).)

Solution

Part (a): True

Official hint. The hints sheet gives only the answer *True*, with no justification.

Idea. (*Scratch work – not part of the written answer.*) Try the smallest possible proper subsequence: delete just the first term. Its index sequence is $n_k = k + 1$, which is strictly increasing and satisfies $n_1 = 2 \neq 1$, so it is proper by the course definition (some k with $n_k \neq k$ suffices; here $k = 1$ already works). If this single proper subsequence converging already forces $\{a_n\}$ to converge, no second subsequence is needed. We also record the more elaborate “two overlapping proper subsequences” route as Method 2, since it is the same overlap technique reused in Question 4.

Method 1: Deleting the first term.

Proof. Define $b_k := a_{k+1}$ for $k \in \mathbb{N}$. The index sequence $n_k = k + 1$ satisfies $n_1 < n_2 < n_3 < \dots$, so $\{b_k\} = \{a_{n_k}\}$ is a subsequence of $\{a_n\}$; moreover $n_1 = 2 \neq 1$, so it is a *proper* subsequence. By hypothesis, every proper subsequence of $\{a_n\}$ converges, so $\{b_k\}$ converges; let $L := \lim_{k \rightarrow \infty} b_k$.

We claim $\{a_n\}$ converges to L . Let $\varepsilon > 0$ be given. Since $b_k \rightarrow L$, there exists $K \in \mathbb{N}$ such that for all $k \in \mathbb{N}$, $k \geq K$ implies $|b_k - L| < \varepsilon$. Set $N := K + 1$. For any $n \in \mathbb{N}$ with $n \geq N$, write $n = k + 1$ where $k = n - 1 \geq N - 1 = K$; then $|a_n - L| = |b_k - L| < \varepsilon$. Therefore, for all $n \geq N$, $|a_n - L| < \varepsilon$, which is exactly the definition of $a_n \rightarrow L$. Hence $\{a_n\}$ converges. $\square$

Method 2: Two overlapping proper subsequences.

Idea. (*Scratch work – not part of the written answer.*) Take two proper subsequences that omit different single indices; they share a common further subsequence, so the Subsequential Limits theorem and Uniqueness of Limits force them to agree.

Proof. Let P denote the subsequence of $\{a_n\}$ with index set $\mathbb{N} \setminus \{1\} = \{2, 3, 4, \dots\}$, and let Q denote the subsequence with index set $\mathbb{N} \setminus \{2\} = \{1, 3, 4, 5, \dots\}$. Both index sets are strictly increasing enumerations of infinite subsets of $\mathbb{N}$ that each omit at least one index

(1 for P , 2 for Q), so P and Q are both proper subsequences of $\{a_n\}$. By hypothesis, P and Q converge; write $P \rightarrow L_P$ and $Q \rightarrow L_Q$.

The index set $R := \{3, 4, 5, \dots\} = \mathbb{N} \setminus \{1, 2\}$ gives a subsequence that is a further subsequence of both P and Q (since $R \subseteq \mathbb{N} \setminus \{1\}$ and $R \subseteq \mathbb{N} \setminus \{2\}$). By the Theorem on Subsequential Limits applied to $P \rightarrow L_P$, this further subsequence converges to L_P ; applied to $Q \rightarrow L_Q$, the same further subsequence converges to L_Q . Since limits are unique, $L_P = L_Q =: L$.

Finally, every index $n \geq 2$ lies in P 's index set, so (exactly as in Method 1, applied to P in place of $\{b_k\}$) the convergence $P \rightarrow L$ already forces $a_n \rightarrow L$. $\square$

Common slip. Forgetting that a subsequence obtained by deleting only *one* term of $\{a_n\}$ is still genuinely proper – the definition requires $n_k \neq k$ for *some* k , not for all k ; some solutions wrongly assume a proper subsequence must omit infinitely many terms.

Beyond the hints. The hints sheet gives no method for (a). Method 1 is the short route the hints sheet does not mention; Method 2 reconstructs the more elaborate “two overlapping proper subsequences” argument that the digest’s gap analysis attributes to the intended solution, and it is the same overlap technique used again in Question 4.

Part (b): False

Official hint. Answer only: False.

Idea. (*Scratch work – not part of the written answer.*) Use the classic counterexample $a_n = n$ for odd n , $a_n = 0$ for even n . Every convergent subsequence is forced to be eventually all-even (else it is unbounded), hence converges only to 0; yet the whole sequence is unbounded, hence divergent.

Counterexample. Define $a_n := n$ if n is odd, and $a_n := 0$ if n is even.

Proof that this is a counterexample. We first show every convergent subsequence of $\{a_n\}$ converges to 0. Let $\{a_{n_k}\}$ be any convergent subsequence, with limit a . Recall the Boundedness of Convergent Sequences: every convergent sequence is bounded. If $\{a_{n_k}\}$ contained infinitely many odd-indexed terms, then, since $n_k \rightarrow \infty$, those terms $a_{n_k} = n_k$ would be unbounded, contradicting boundedness. Therefore $\{a_{n_k}\}$ has only finitely many odd-indexed terms; that is, all but finitely many of its terms are even-indexed, hence equal to 0. So $\{a_{n_k}\}$ is eventually the constant sequence 0, and therefore $a = 0$.

Thus every convergent subsequence of $\{a_n\}$ converges to 0; in particular, if $\{a_n\}$ itself converged (it is trivially a subsequence of itself, with $n_k = k$), it would have to converge to 0. However $\{a_n\} \not\rightarrow 0$: taking $\varepsilon = 1$, for every $N \in \mathbb{N}$, by the Archimedean Property there is an odd $n \geq N$ with $a_n = n \geq N > 1 = \varepsilon$, so the definition of $a_n \rightarrow 0$ fails. Hence $\{a_n\}$ diverges, while every convergent subsequence converges to 0. This falsifies the statement. $\square$

Common slip. Concluding “ $\{a_n\}$ converges to a ” merely because a is the unique value any convergent subsequence can reach; convergence of the subsequences says nothing about the (unbounded, divergent) sequence as a whole.

Beyond the hints. The hints sheet gives no counterexample. We identify one, prove every convergent subsequence is forced to 0 (via Boundedness of Convergent Sequences applied to the odd-indexed terms), and give an explicit argument that $\{a_n\} \not\rightarrow 0$.

Part (c): True

Official hint. Answer only: True.

Idea. (*Scratch work – not part of the written answer.*) Contrapositive: if $\{a_n\}$ converges, Subsequential Limits forces every subsequence, including the given divergent one, to converge too – a contradiction.

Proof (by contrapositive). We prove the contrapositive: if $\{a_n\}$ converges, then every subsequence of $\{a_n\}$ converges (so $\{a_n\}$ cannot contain a divergent subsequence). Suppose $\{a_n\} \rightarrow a$ for some $a \in \mathbb{R}$, and let $\{a_{n_k}\}$ be an arbitrary subsequence. By the Theorem on Subsequential Limits, since $\{a_n\} \rightarrow a$, the subsequence $\{a_{n_k}\} \rightarrow a$ as well; in particular it converges. As $\{a_{n_k}\}$ was an arbitrary subsequence, every subsequence of $\{a_n\}$ converges.

This is precisely the contrapositive of “if $\{a_n\}$ contains a divergent subsequence, then $\{a_n\}$ diverges.” Since the contrapositive holds, so does the original statement. $\square$

Common slip. Trying to prove this directly from the ε - N definition instead of citing Subsequential Limits by name; the contrapositive route is immediate once the theorem is cited.

Beyond the hints. Spells out the contrapositive logic and names the cited theorem explicitly (the hints sheet gives only “True”).

Question 2 (True/False – Existence of Sequences with Prescribed Subsequences)

Problem

Determine if each of the following statements is true.

- (a) There is a sequence that does not contain 0 or 1 as a term but contains subsequences converging to each of these values.
- (b) There is a sequence with a subsequence that is bounded but contains no subsequence that converges.
- (c) There is a monotone sequence that contains two subsequences converging to different limits.

(Tutorial 3, Question 2; Lectures 7–9. Part (b) is also Sample Midterm Question 1(b).)

Solution

Part (a): True

Official hint. Answer only: True.

Idea. (Scratch work – not part of the written answer.) Interleave $1/2, 1/3, 1/4, \dots \rightarrow 0$ with $1 - 1/2, 1 - 1/3, \dots \rightarrow 1$; neither family ever hits 0 or 1, since $1/n \in (0, 1)$ once $n \geq 2$.

Construction. For $n \in \mathbb{N}$ define $c_n := \frac{1}{n+1}$ and $d_n := 1 - \frac{1}{n+1}$, and interleave them as

$$a_{2n-1} := c_n, \quad a_{2n} := d_n \quad (n \in \mathbb{N}),$$

so that $\{a_n\} = c_1, d_1, c_2, d_2, c_3, d_3, \dots = \frac{1}{2}, \frac{1}{2}, \frac{1}{3}, \frac{2}{3}, \frac{1}{4}, \frac{3}{4}, \dots$

Proof. For every $n \in \mathbb{N}$, $n+1 \geq 2$, so $0 < \frac{1}{n+1} \leq \frac{1}{2} < 1$; hence $c_n \in (0, 1)$ and $d_n = 1 - c_n \in (0, 1)$ as well. Therefore no term of $\{a_n\}$ equals 0 or 1.

The sequence $c_n = 1/(n+1) \rightarrow 0$ (a standard limit, via the Archimedean Property). By the Algebraic Limit Theorem, applied to the constant sequence 1 and $c_n \rightarrow 0$, $d_n = 1 - c_n \rightarrow 1 - 0 = 1$. The subsequence $\{a_{2n-1}\} = \{c_n\}$ of $\{a_n\}$ converges to 0, and the subsequence $\{a_{2n}\} = \{d_n\}$ converges to 1. So $\{a_n\}$ contains subsequences converging to each of 0 and 1, while never taking either value as a term. $\square$

Common slip. Using $c_n = 1/n$ starting at $n = 1$ without shifting the index – then $c_1 = 1$, one of the two forbidden values.

Beyond the hints. Supplies the explicit interleaved formula and the index shift ($n+1$, not n) needed to avoid the forbidden values, which the answer-only hints sheet omits.

Part (b): False

Official hint. Answer only: False.

Idea. (*Scratch work – not part of the written answer.*) Apply Bolzano-Weierstrass to the bounded subsequence itself, treated as a sequence in its own right; a subsequence of a subsequence is a subsequence of the original.

Proof. Suppose, for contradiction, that some sequence $\{a_n\}$ has a subsequence $\{a_{n_k}\}$ that is bounded, while $\{a_n\}$ has no convergent subsequence at all.

Regard $\{a_{n_k}\}_{k=1}^{\infty}$ as a sequence in its own right (indexed by k); it is bounded by hypothesis. By the Bolzano-Weierstrass Theorem, $\{a_{n_k}\}$ has a convergent subsequence $\{a_{n_{k_j}}\}_{j=1}^{\infty}$, where $k_1 < k_2 < k_3 < \dots$ is a strictly increasing sequence of natural numbers.

We claim $\{a_{n_{k_j}}\}_j$ is a subsequence of $\{a_n\}$. Since $\{n_k\}$ is strictly increasing in k and $\{k_j\}$ is strictly increasing in j , the composite index map $j \mapsto n_{k_j}$ is strictly increasing in j (a strictly increasing function of a strictly increasing function is strictly increasing). Hence $n_{k_1} < n_{k_2} < n_{k_3} < \dots$ is a valid strictly increasing sequence of natural numbers, so $\{a_{n_{k_j}}\}_j$ is indeed a subsequence of the original sequence $\{a_n\}$, and it converges.

This contradicts the assumption that $\{a_n\}$ has no convergent subsequence. Therefore no such sequence exists, and the statement is false. $\square$

Common slip. Not checking that “a subsequence of a subsequence” is a genuine subsequence of the original sequence; this requires the composition-of-strictly-increasing-index-maps fact (Lecture 7’s Functional View remark), not just an appeal to intuition.

Beyond the hints. The hints sheet does not cite the Bolzano-Weierstrass Theorem at all for this part; we make the citation explicit and justify the composition step the digest flags as missing.

Part (c): False

Official hint. Answer only: False.

Idea. (*Scratch work – not part of the written answer.*) Split into the bounded case (Monotone Convergence Theorem + Subsequential Limits) and the unbounded case (every subsequence is itself monotone and unbounded, hence no subsequence converges at all).

Proof. Let $\{a_n\}$ be monotone; without loss of generality suppose it is increasing (the decreasing case is symmetric, e.g. by applying the increasing case to $\{-a_n\}$).

Case 1: $\{a_n\}$ is bounded. Since $\{a_n\}$ is monotone and bounded, the Monotone Convergence Theorem gives that $\{a_n\}$ converges, to some limit L . By the Theorem on Subsequential Limits, every subsequence of $\{a_n\}$ converges to L as well. In particular, any two convergent subsequences of $\{a_n\}$ must both converge to L ; they cannot converge to different limits.

Case 2: $\{a_n\}$ is unbounded (above, since it is increasing). We show every subsequence of $\{a_n\}$ is also increasing and unbounded, hence no subsequence of $\{a_n\}$ converges at all (so there certainly cannot be two subsequences converging to different limits). Let $\{a_{n_k}\}$ be any subsequence. Since $n_1 < n_2 < \dots$ are indices of the increasing sequence $\{a_n\}$, monotonicity gives $a_{n_1} \leq a_{n_2} \leq a_{n_3} \leq \dots$, so $\{a_{n_k}\}$ is increasing. Given any $M \in \mathbb{R}$, since $\{a_n\}$ is unbounded above and increasing, there is $N \in \mathbb{N}$ with $a_N > M$, and $a_n \geq a_N > M$ for all $n \geq N$. Since $n_k \rightarrow \infty$, there is $K \in \mathbb{N}$ with $n_K \geq N$, so $a_{n_k} \geq a_{n_K} > M$ for all $k \geq K$. Hence $\{a_{n_k}\}$ is also unbounded above. By the Boundedness of Convergent

Sequences, an unbounded sequence cannot converge, so $\{a_{n_k}\}$ diverges.

In both cases, $\{a_n\}$ cannot have two subsequences converging to different limits. Hence the statement is false. $\square$

Common slip. Assuming “monotone” automatically means “bounded” – the two are independent hypotheses; the unbounded case needs its own argument, since the Monotone Convergence Theorem does not apply without boundedness.

Beyond the hints. Supplies both the bounded case (via MCT + Subsequential Limits, as the digest’s gap analysis calls for) and the unbounded case (which the digest notes is entirely missing from the hints), including the explicit unboundedness-transfer argument for subsequences.

Question 3 (Shuffled Sequences and the Divergence Criterion)

Problem

In this question, $\{a_n\}$ and $\{b_n\}$ are two given convergent sequences, and $\{c_n\}$ denotes the “shuffled” sequence

$$a_1, b_1, a_2, b_2, a_3, b_3, \dots, a_n, b_n, \dots$$

where the subsequence indexed by odd numbers coincide with $\{a_n\}$, and the subsequence indexed by even numbers coincide with $\{b_n\}$.

Use the Divergence Criterion to prove that if $\{a_n\}$ and $\{b_n\}$ have the same limit, then $\{c_n\}$ is convergent. **Do not use the definition of limits of sequences.** (*Tutorial 3, Question 3; Lectures 7–9.*)

Solution

Method 1: Contrapositive of the (bounded) Divergence Criterion

Official hint. (a) Since the Divergence Criterion is for divergent sequences, try proving the contrapositive by applying it to $\{c_n\}$ to obtain two subsequences $\{c_{n_k}\}$ and $\{c_{m_k}\}$. (b) Show that any subsequence $\{c_{n_k}\}$ of $\{c_n\}$ must have infinitely many terms appearing in either $\{a_n\}$ or $\{b_n\}$, and hence shares a common subsequence with either $\{a_n\}$ or $\{b_n\}$. Apply this to the two subsequences above and relate their limits with the limits of $\{a_n\}$ and $\{b_n\}$.

Idea. (*Scratch work – not part of the written answer.*) The Divergence Criterion applies only to *bounded* sequences, so boundedness of $\{c_n\}$ must be established first, from boundedness of the convergent sequences $\{a_n\}, \{b_n\}$. The Criterion says a bounded sequence is divergent iff it has two subsequences with different limits, so it suffices to rule out that possibility; the cleanest way is to show *every* convergent subsequence of $\{c_n\}$ converges to the common limit $L := \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} b_n$, via a pigeonhole argument on odd/even indices.

Let $L := \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} b_n$.

Step 1: $\{c_n\}$ is bounded. Since $\{a_n\}$ and $\{b_n\}$ converge, the Boundedness of Convergent Sequences gives $M_a, M_b > 0$ with $|a_n| \leq M_a$ and $|b_n| \leq M_b$ for all $n \in \mathbb{N}$. Every term of $\{c_n\}$ equals some a_m (odd index) or some b_m (even index), so with $M := \max\{M_a, M_b\}$ we get $|c_n| \leq M$ for all $n \in \mathbb{N}$. Hence $\{c_n\}$ is bounded.

Step 2: every convergent subsequence of $\{c_n\}$ converges to L . Let $\{c_{n_k}\}_{k=1}^\infty$ be an arbitrary subsequence of $\{c_n\}$ that converges, say to L' . Partition $\mathbb{N}$ into the odd indices O and even indices E ; both are infinite, and $O \cup E = \mathbb{N}$. Since $\{n_k : k \in \mathbb{N}\}$ is an infinite subset of $\mathbb{N} = O \cup E$, at least one of $\{n_k : k \in \mathbb{N}\} \cap O$ and $\{n_k : k \in \mathbb{N}\} \cap E$ is infinite (otherwise both intersections, hence their union $\{n_k : k \in \mathbb{N}\}$, would be finite – a contradiction).

Case A: $\{n_k : k \in \mathbb{N}\} \cap O$ is infinite. List the indices k for which n_k is odd, in increasing order, as $k_1 < k_2 < k_3 < \dots$; this is a strictly increasing sequence of natural numbers, since it is a subset of $\{1, 2, 3, \dots\}$ listed in increasing order. Then $\left\{c_{n_{k_j}}\right\}_{j=1}^\infty$ is a

subsequence of $\{c_{n_k}\}_k$, so by the Theorem on Subsequential Limits applied to $\{c_{n_k}\} \rightarrow L'$, we get $c_{n_{k_j}} \rightarrow L'$.

On the other hand, since each n_{k_j} is odd, write $n_{k_j} = 2m_j - 1$ for some $m_j \in \mathbb{N}$; then $c_{n_{k_j}} = a_{m_j}$ by the definition of the shuffled sequence. Because $j \mapsto n_{k_j}$ is strictly increasing and $n_{k_j} = 2m_j - 1$, the map $j \mapsto m_j$ is also strictly increasing, so $\{a_{m_j}\}_{j=1}^\infty$ is a genuine subsequence of $\{a_n\}$. Since $\{a_n\} \rightarrow L$, the Theorem on Subsequential Limits gives $a_{m_j} \rightarrow L$, i.e. $c_{n_{k_j}} \rightarrow L$.

We have shown $c_{n_{k_j}} \rightarrow L'$ (as a subsequence of $\{c_{n_k}\}$) and $c_{n_{k_j}} \rightarrow L$ (as a subsequence of $\{a_n\}$, relabelled). By the Uniqueness of Limits, $L' = L$.

Case B: $\{n_k : k \in \mathbb{N}\} \cap E$ is infinite. By the identical argument with $\{b_n\}$ in place of $\{a_n\}$ (even indices $n_{k_j} = 2m_j$ give $c_{n_{k_j}} = b_{m_j}$, and $\{b_n\} \rightarrow L$), we again conclude $L' = L$.

In either case $L' = L$. Since $\{c_{n_k}\}$ was an arbitrary convergent subsequence of $\{c_n\}$, every convergent subsequence of $\{c_n\}$ converges to L .

Step 3: no two subsequences of $\{c_n\}$ converge to different limits. By Step 2, any two convergent subsequences of $\{c_n\}$ both converge to L , so they cannot converge to different limits.

Step 4: conclude via the Divergence Criterion. By Step 1, $\{c_n\}$ is bounded. The Divergence Criterion states that a bounded sequence is divergent if and only if it has two subsequences converging to different limits. By Step 3, $\{c_n\}$ has no such pair of subsequences. Therefore, by the contrapositive of the “only if” direction, $\{c_n\}$ is not divergent. Since every sequence is, by definition, either convergent or divergent, $\{c_n\}$ must be convergent. $\square$

(In fact Step 2, applied to $\{c_n\}$ itself – which is trivially a subsequence of itself with $n_k = k$ – already shows that if $\{c_n\}$ converges, its limit is L ; combined with Step 4 this pins down $\lim c_n = L = \lim a_n = \lim b_n$.)

Worked example. Take $a_n := 1/n \rightarrow 0$ and $b_n := -1/n \rightarrow 0$, so $L = 0$. The shuffled sequence is

$$c_n = 1, -1, \frac{1}{2}, -\frac{1}{2}, \frac{1}{3}, -\frac{1}{3}, \dots,$$

which oscillates in sign yet visibly shrinks towards 0; Method 1 confirms $c_n \rightarrow 0$ without ever writing an ε - N argument for $\{c_n\}$ directly. Contrast this with $a_n := 1/n \rightarrow 0$ and $b_n := 1 + 1/n \rightarrow 1$ (different limits, so the hypothesis of the question fails): here $\{c_n\} = 1, 2, \frac{1}{2}, \frac{3}{2}, \frac{1}{3}, \frac{4}{3}, \dots$ has the two subsequences $\{a_n\} \rightarrow 0$ and $\{b_n\} \rightarrow 1$ sitting inside it with different limits, so by the Divergence Criterion $\{c_n\}$ diverges – illustrating exactly why the “same limit” hypothesis cannot be dropped, and why the whole argument hinges on Step 2 forcing every convergent subsequence to the single value L .

Common slip. Applying the Divergence Criterion without first checking boundedness – the course’s Divergence Criterion (Lecture 7, deck p. 14) is stated only for *bounded* sequences, and it is exactly the “only if” direction (used here by contrapositive) that needs this hypothesis. Another common slip is treating “every subsequence has infinitely many odd or even indices” as obvious without invoking the partition/pigeonhole argument, and forgetting that the reindexing $n_{k_j} = 2m_j - 1 \mapsto m_j$ must itself be shown strictly increasing before citing Subsequential Limits on $\{a_n\}$.

Beyond the hints. The hints give only the two-step strategy sketch; we supply: the boundedness argument (Step 1, entirely absent from the hints); the precise pigeonhole statement that an infinite index set split into two pieces has an infinite piece (Step 2);

the explicit reindexing $n_{k_j} = 2m_j - 1$ (odd case) / $n_{k_j} = 2m_j$ (even case) that identifies the “common subsequence” the hint refers to; and the final assembly via the convergent/divergent dichotomy (Step 4) – all without appealing to the ε - N definition of a limit, as required.

$$\boxed{\{c_n\} \text{ converges, to } L = \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} b_n.}$$

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Question 4 (Three Overlapping Subsequences)

Problem

Prove that if $\{a_n\}$ is a sequence for which the subsequences $\{a_{2k}\}$, $\{a_{2k+1}\}$, and $\{a_{3k}\}$ are convergent, then it is convergent. (*Tutorial 3, Question 4; Lectures 7–9; builds on the shuffled-sequence technique of Question 3.*)

Solution

Method 1: Overlapping the three subsequences, then adapting Question 3

Official hint. (a) Find a subsequence of $\{a_{2k}\}$ and of $\{a_{3k}\}$ that coincide. Then use the theorem on the Subsequential Limits to compare the limits of $\{a_{2k}\}$ and $\{a_{3k}\}$. (b) The same comparison of limits holds for $\{a_{2k+1}\}$ and $\{a_{3k}\}$. (c) The preceding question can be adapted to complete the proof.

Idea. (*Scratch work – not part of the written answer.*) Multiples of 6 lie in both $\{a_{2k}\}$ (as $6k = 2(3k)$) and $\{a_{3k}\}$ (as $6k = 3(2k)$), forcing their limits to agree; likewise $6k + 3$ lies in both $\{a_{2k+1}\}$ and $\{a_{3k}\}$. This gives one common limit L for all three. Then, since every index is even or odd, the odd/even partition argument of Question 3 adapts directly, using $\{a_{2k}\} \rightarrow L$ and $\{a_{2k+1}\} \rightarrow L$ in place of $\{a_n\}, \{b_n\}$.

Let $L_1 := \lim_{k \rightarrow \infty} a_{2k}$, $L_2 := \lim_{k \rightarrow \infty} a_{2k+1}$, $L_3 := \lim_{k \rightarrow \infty} a_{3k}$.

Step 1: $L_1 = L_3$. For $k \in \mathbb{N}$, the index $6k$ satisfies $6k = 2(3k)$, so $6k$ is the $(3k)$ -th term of the index family $\{2j\}_{j \in \mathbb{N}}$ (take $j = 3k$); since $j = 3k$ is strictly increasing in k , $\{a_{6k}\}_{k=1}^\infty$ is a subsequence of $\{a_{2j}\}_{j=1}^\infty$. By the Theorem on Subsequential Limits, since $\{a_{2j}\} \rightarrow L_1$, we get $a_{6k} \rightarrow L_1$.

Likewise $6k = 3(2k)$, so $6k$ is the $(2k)$ -th term of $\{3j\}_{j \in \mathbb{N}}$; since $j = 2k$ is strictly increasing in k , $\{a_{6k}\}_k$ is also a subsequence of $\{a_{3j}\}_j$, so $a_{6k} \rightarrow L_3$.

By the Uniqueness of Limits, $L_1 = L_3$.

Step 2: $L_2 = L_3$. For $k \in \mathbb{N}$, the index $6k + 3$ satisfies $6k + 3 = 2(3k + 1) + 1$, so a_{6k+3} is the term of $\{a_{2j+1}\}_{j=1}^\infty$ with index $j = 3k + 1 \geq 4$, and $j = 3k + 1$ is strictly increasing in k ; hence $\{a_{6k+3}\}_k$ is a subsequence of $\{a_{2j+1}\}_j$, so $a_{6k+3} \rightarrow L_2$.

Also $6k + 3 = 3(2k + 1)$, the $(2k + 1)$ -th term of $\{3j\}_{j \in \mathbb{N}}$ (with $j = 2k + 1$, strictly increasing in k), so $\{a_{6k+3}\}_k$ is a subsequence of $\{a_{3j}\}_j \rightarrow L_3$, giving $a_{6k+3} \rightarrow L_3$.

By the Uniqueness of Limits, $L_2 = L_3$.

Combining Steps 1–2, $L_1 = L_2 = L_3 =: L$.

Step 3: every convergent subsequence of $\{a_n\}$ converges to L . This is the adapted version of Question 3's Step 2, with $\mathbb{N}$ split into evens and odds. Let $\{a_{n_k}\}_k$ be an arbitrary convergent subsequence of $\{a_n\}$, with limit L' . Since $\{n_k : k \in \mathbb{N}\}$ is an infinite subset of $\mathbb{N} = \{\text{evens}\} \cup \{\text{odds}\}$, at least one of its intersections with the evens or the odds is infinite.

If the intersection with the evens is infinite: exactly as in Question 3 (Case A, with $\{a_n\}$ there replaced by $\{a_{2m}\}$ here), extracting those terms gives a further subsequence that is a subsequence of both $\{a_{n_k}\}$ (so it converges to L') and of $\{a_{2m}\}$ (so it converges to $L_1 = L$). By Uniqueness of Limits, $L' = L$.

If the intersection with the odds is infinite: symmetrically, using $\{a_{2m+1}\} \rightarrow L_2 = L$, we again get $L' = L$.

Either way, $L' = L$.

Step 4: $\{a_n\}$ is bounded. Since $\{a_{2k}\}$ and $\{a_{2k+1}\}$ both converge, the Boundedness of Convergent Sequences gives $M_1, M_2 > 0$ with $|a_{2k}| \leq M_1$ and $|a_{2k+1}| \leq M_2$ for all $k \in \mathbb{N}$. Every $n \in \mathbb{N}$ is either even or odd, so with $M := \max\{M_1, M_2\}$, $|a_n| \leq M$ for all $n \in \mathbb{N}$. (Note that $\{a_{3k}\}$ is not needed for boundedness, since the evens and odds alone already exhaust $\mathbb{N}$.)

Proof (assembly). By Step 4, $\{a_n\}$ is bounded. By Step 3, any two convergent subsequences of $\{a_n\}$ both converge to L , so $\{a_n\}$ has no two subsequences converging to different limits. By the Divergence Criterion (contrapositive of the “only if” direction, as in Question 3, Step 4), $\{a_n\}$ is therefore not divergent, hence convergent (by the convergent/divergent dichotomy). $\square$

Why all three subsequences are needed. The hypothesis on $\{a_{2k}\}$ and $\{a_{2k+1}\}$ alone is not enough: define $a_n := 0$ for n even and $a_n := 1$ for n odd. Then $\{a_{2k}\} \equiv 0 \rightarrow 0$ and $\{a_{2k+1}\} \equiv 1 \rightarrow 1$ both converge, yet $\{a_n\} = 0, 1, 0, 1, \dots$ is bounded with two subsequences converging to different limits (0 and 1), so by the Divergence Criterion $\{a_n\}$ itself diverges. (Consistently, $\{a_{3k}\}$ here is $a_3, a_6, a_9, \dots = 1, 0, 1, 0, \dots$, which also diverges – so this example does not satisfy the question’s hypotheses, exactly because the third subsequence is what is missing.) The role of the hypothesis on $\{a_{3k}\}$ is precisely to rule this scenario out: Steps 1–2 use it to force $L_1 = L_2$, which is the one fact the two-subsequence hypothesis alone cannot supply.

Common slip. Trying to build a common subsequence directly from $\{a_{2k+1}\}$ and $\{a_{2k}\}$ (they are disjoint in index and share no common further subsequence); the correct pairings are $(\{a_{2k}\}, \{a_{3k}\})$ via multiples of 6, and $(\{a_{2k+1}\}, \{a_{3k}\})$ via $6k+3$ – both routed *through* $\{a_{3k}\}$, which is exactly why the problem supplies that third subsequence.

Beyond the hints. The hints name the two comparisons “ $\{a_{2k}\}$ vs. $\{a_{3k}\}$ ” and “ $\{a_{2k+1}\}$ vs. $\{a_{3k}\}$ ” but do not give the concrete shared indices ($6k$ and $6k+3$ respectively); we supply both explicitly, together with the strictly-increasing reindexing that legitimises each as a subsequence. We also spell out precisely how “the preceding question adapts” (Step 3, mirroring Question 3’s pigeonhole argument with the covering $\mathbb{N} = \text{evens} \cup \text{odds}$ in place of $\mathbb{N} = O \cup E$) and the boundedness step, which the hints omit entirely.

$$\boxed{\{a_n\} \text{ converges, to } L = \lim_{k \rightarrow \infty} a_{2k} = \lim_{k \rightarrow \infty} a_{2k+1} = \lim_{k \rightarrow \infty} a_{3k}.$$

Question 5 (Constructing a Subsequence Converging to Zero)

Problem

Prove that if $\{a_n\}$ is a sequence such that for every positive number $\varepsilon > 0$, there are infinitely many terms satisfying $|a_n| < \varepsilon$, then it has a subsequence $\{a_{n_k}\}$ converging to 0. (*Tutorial 3, Question 5; Lectures 4–6 and 7–9.*)

Solution

Method 1: Recursive construction with $\varepsilon_k = 1/k$

Official hint. (a) Use an arbitrary, but fixed, sequence $\{\varepsilon_k\}$ of positive real numbers converging to 0 to construct the subsequence. (b) For each $k \in \mathbb{N}$, pick the next term $a_{n_{k+1}}$ to satisfy $|a_{n_{k+1}}| < \varepsilon_k$, with $n_{k+1} > n_k$. (c) Use the definition of limits of sequences to prove the constructed subsequence converges to 0.

Idea. (*Scratch work – not part of the written answer.*) Take the concrete choice $\varepsilon_k = 1/k$. At each stage the hypothesis gives infinitely many indices n with $|a_n| < \varepsilon_k$; since only finitely many indices are $\leq n_{k-1}$, infinitely many of these qualifying indices still lie strictly beyond n_{k-1} , so one can always be chosen for n_k . This “infinitely many minus finitely many is still infinite” step is the subtle point the hints sheet skips.

Step 1: construction. For each $k \in \mathbb{N}$, let $\varepsilon_k := 1/k$; then $\varepsilon_k > 0$ for all k and $\varepsilon_k \rightarrow 0$ (Archimedean Property). We construct $n_1 < n_2 < n_3 < \dots$ recursively.

Base case. By hypothesis (with $\varepsilon = \varepsilon_1 = 1$), the set $S_1 := \{n \in \mathbb{N} : |a_n| < 1\}$ is infinite, in particular nonempty. By the Well-Ordering Principle, S_1 has a least element; let $n_1 := \min S_1$. Then $|a_{n_1}| < \varepsilon_1$.

Inductive step. Suppose $n_1 < n_2 < \dots < n_k$ have been chosen with $|a_{n_j}| < \varepsilon_j$ for $j = 1, \dots, k$. By hypothesis (with $\varepsilon = \varepsilon_{k+1}$), the set $S_{k+1} := \{n \in \mathbb{N} : |a_n| < \varepsilon_{k+1}\}$ is infinite. The set $\{1, 2, \dots, n_k\}$ of indices at most n_k is finite, so $S_{k+1} \cap \{n \in \mathbb{N} : n > n_k\}$ is still infinite: if it were finite, S_{k+1} , being contained in the union of this set and the finite set $\{1, \dots, n_k\}$, would itself be finite – a contradiction. In particular this set is nonempty, so by the Well-Ordering Principle it has a least element; let $n_{k+1} := \min(S_{k+1} \cap \{n > n_k\})$. Then $n_{k+1} > n_k$ and $|a_{n_{k+1}}| < \varepsilon_{k+1}$.

By induction, this defines n_k for every $k \in \mathbb{N}$, with $n_1 < n_2 < n_3 < \dots$ (so $\{a_{n_k}\}$ is a genuine subsequence of $\{a_n\}$) and $|a_{n_k}| < \varepsilon_k = 1/k$ for every $k \in \mathbb{N}$.

Step 2: the constructed subsequence converges to 0.

Proof. Let $\varepsilon > 0$ be given. By the Archimedean Property, there exists $K \in \mathbb{N}$ with $1/K < \varepsilon$. Let $k \in \mathbb{N}$ with $k \geq K$. Since $1/k$ is a decreasing function of k , $k \geq K$ gives $1/k \leq 1/K < \varepsilon$. By Step 1, $|a_{n_k}| < 1/k \leq 1/K < \varepsilon$. Hence, for all $k \geq K$, $|a_{n_k} - 0| < \varepsilon$. Since $\varepsilon > 0$ was arbitrary, this is exactly the definition of $\lim_{k \rightarrow \infty} a_{n_k} = 0$. $\square$

Worked example. Let $a_n := n$ if n is not a perfect square, and $a_n := 1/\sqrt{n}$ if n is a perfect square (so $a_1 = 1, a_4 = \frac{1}{2}, a_9 = \frac{1}{3}, a_{16} = \frac{1}{4}, \dots$, while all other terms are large). For any $\varepsilon > 0$, the terms $a_{k^2} = 1/k$ ($k \in \mathbb{N}$) satisfy $|a_{k^2}| < \varepsilon$ for all k beyond some point, and there are infinitely many perfect squares, so the hypothesis of the question holds.

even though $\{a_n\}$ itself is wildly unbounded and does not converge. The construction of Step 1 extracts exactly the perfect-square terms (or a further thinning of them) as the subsequence $\{a_{n_k}\}$, and Step 2 shows this extracted subsequence converges to 0 – illustrating that the conclusion is genuinely about a *subsequence*, not about $\{a_n\}$ as a whole.

Common slip. Choosing n_{k+1} to merely satisfy $|a_{n_{k+1}}| < \varepsilon_{k+1}$ without checking $n_{k+1} > n_k$ – an index sequence that is not strictly increasing does not define a subsequence at all (Lecture 7’s definition requires $n_1 < n_2 < n_3 < \dots$). Another slip is treating the existence of a qualifying index beyond n_k as “obvious”; it requires the finite/infinite set-difference argument given above.

Beyond the hints. The hints sheet does not name a concrete $\{\varepsilon_k\}$ (we use $\varepsilon_k = 1/k$) and does not justify why the “infinitely many terms” hypothesis is never exhausted by the earlier choices $n_1, \dots, n_k$; we supply the finite-set-removal argument in full, plus the explicit ε - K write-up for Step 2.

Such a subsequence $\{a_{n_k}\}$ exists, with $a_{n_k} \rightarrow 0$, e.g. via $\varepsilon_k = 1/k$.

Question A1 (Limit Superior and Limit Inferior)

Problem

In this question, $\{a_n\}$ denotes a bounded sequence. Recall that the *limit superior* of the sequence $\{a_n\}$ is

$$\limsup_{n \rightarrow \infty} a_n := \lim_{n \rightarrow \infty} \sup\{a_k : k \in \mathbb{N} \text{ and } k \geq n\}$$

and the *limit inferior* of the sequence $\{a_n\}$ is

$$\liminf_{n \rightarrow \infty} a_n := \lim_{n \rightarrow \infty} \inf\{a_k : k \in \mathbb{N} \text{ and } k \geq n\}.$$

Recall also that $\liminf a_n \leq \limsup a_n$.

Prove that $\{a_n\}$ is convergent if and only if $\liminf a_n = \limsup a_n$, and that in this case $\liminf a_n = \limsup a_n = \lim a_n$. (*Tutorial 3, Additional Question 1; Lectures 2, 5, 7–9.*)

Solution

Method 1: Monotone Convergence Theorem + Squeeze Theorem + Order Limit Theorem

Official hint. (a) For “if”, compare a_n with the corresponding terms of $\{\sup_{k \geq n} a_k\}$ and $\{\inf_{k \geq n} a_k\}$, then apply the Squeeze Theorem. (b) For “only if”, use the definition of limit to bound tails, then bound the sup/inf sequences via the definitions of suprema/infima, and apply the Order Limit Theorem.

Idea. (*Scratch work – not part of the written answer.*) Before either direction, $\limsup$ / $\liminf$ must be shown to exist as genuine limits: define $u_n := \sup\{a_k : k \geq n\}$, $l_n := \inf\{a_k : k \geq n\}$; these are well-defined (Axiom of Completeness, since $\{a_n\}$ is bounded), monotone, and bounded, so the Monotone Convergence Theorem gives that $\lim u_n$ and $\lim l_n$ exist – this is exactly what “ $\limsup a_n$ ” and “ $\liminf a_n$ ” mean.

Step 0: $\{u_n\}$ and $\{l_n\}$ are well-defined, monotone, and convergent. Since $\{a_n\}$ is bounded, there is $M > 0$ with $|a_n| \leq M$ for all n . For each $n \in \mathbb{N}$, the tail set $T_n := \{a_k : k \in \mathbb{N}, k \geq n\}$ is nonempty and bounded above by M (and below by $-M$), so by the Axiom of Completeness, $u_n := \sup T_n$ and $l_n := \inf T_n$ both exist, with $-M \leq l_n \leq u_n \leq M$.

For $n \in \mathbb{N}$, $T_{n+1} \subseteq T_n$ (the tail from $n+1$ is a subset of the tail from n), so every upper bound for T_n is an upper bound for T_{n+1} ; in particular u_n is an upper bound for T_{n+1} , so $u_{n+1} = \sup T_{n+1} \leq u_n$. Thus $\{u_n\}$ is decreasing. Symmetrically, $\{l_n\}$ is increasing. Both are bounded (by $\pm M$), so the Monotone Convergence Theorem gives that $\{u_n\}$ and $\{l_n\}$ converge; write $U := \limsup a_n = \lim u_n$ and $Li := \liminf a_n = \lim l_n$.

Also, since $a_n \in T_n$ for every n (as $n \geq n$), $l_n \leq a_n \leq u_n$ for every $n \in \mathbb{N}$.

Step 1 (“if”): $Li = U =: L \implies a_n \rightarrow L$.

Proof. Suppose $Li = U =: L$. By Step 0, $l_n \leq a_n \leq u_n$ for every $n \in \mathbb{N}$, and $l_n \rightarrow L$, $u_n \rightarrow L$. By the Squeeze Theorem, $a_n \rightarrow L$. Hence $\{a_n\}$ converges, and $\lim a_n = L = \liminf a_n = \limsup a_n$. $\square$

Step 2 (“only if”): $a_n \rightarrow a \implies \text{Li} = \text{U} = a$.

Proof. Suppose $a_n \rightarrow a$. Let $\varepsilon > 0$. By the definition of $a = \lim a_n$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$, $a - \frac{\varepsilon}{2} < a_n < a + \frac{\varepsilon}{2}$.

Fix $n \geq N$. Every element of the tail set $T_n = \{a_k : k \geq n\}$ has index $k \geq n \geq N$, so every $a_k \in T_n$ satisfies $a - \frac{\varepsilon}{2} < a_k < a + \frac{\varepsilon}{2}$. Thus $a + \frac{\varepsilon}{2}$ is an upper bound for T_n , so by the definition of supremum, $u_n = \sup T_n \leq a + \frac{\varepsilon}{2}$. Also $a_n \in T_n$ satisfies $a_n > a - \frac{\varepsilon}{2}$, and $u_n \geq a_n$ (the supremum is at least any element), so $u_n > a - \frac{\varepsilon}{2}$. Hence, for all $n \geq N$,

$$a - \frac{\varepsilon}{2} < u_n \leq a + \frac{\varepsilon}{2},$$

i.e., $|u_n - a| \leq \frac{\varepsilon}{2} < \varepsilon$. Symmetrically, $a - \frac{\varepsilon}{2}$ is a lower bound for T_n , so $l_n = \inf T_n \geq a - \frac{\varepsilon}{2}$, and $l_n \leq a_n < a + \frac{\varepsilon}{2}$, giving $|l_n - a| \leq \frac{\varepsilon}{2} < \varepsilon$ for all $n \geq N$.

Since $\varepsilon > 0$ was arbitrary, this shows $u_n \rightarrow a$ and $l_n \rightarrow a$. Since $\{u_n\}$ and $\{l_n\}$ were already known (Step 0) to converge to U and Li respectively, and limits are unique, it therefore follows that $\text{U} = a = \text{Li}$. $\square$

Worked example. Let $a_n := (-1)^n$, so $\{a_n\} = -1, 1, -1, 1, \dots$ is bounded but divergent. For each n , $T_n = \{a_k : k \geq n\} = \{-1, 1\}$, so $u_n = \sup T_n = 1$ and $l_n = \inf T_n = -1$ for every n ; hence $\limsup a_n = \lim u_n = 1$ and $\liminf a_n = \lim l_n = -1$. Since $1 \neq -1$, Step 2’s contrapositive (equivalently, the “only if” direction read as “not equal $\implies$ not convergent”) correctly predicts $\{a_n\}$ diverges. By contrast, for $a_n := (-1)^n/n \rightarrow 0$, the tail sets T_n shrink towards $\{0\}$ as $n \rightarrow \infty$, so both $u_n \rightarrow 0$ and $l_n \rightarrow 0$; hence $\limsup a_n = \liminf a_n = 0 = \lim a_n$, consistent with Step 1.

Common slip. Treating $\limsup a_n$ and $\liminf a_n$ as though they were “obviously” limits without first proving $\{u_n\}, \{l_n\}$ converge (Step 0) – until the Monotone Convergence Theorem is invoked, “the limit of the sup sequence” is not automatically meaningful. A second slip is using strict inequalities throughout and then trying to conclude a strict bound on u_n or l_n ; since sup/inf need not be attained, only $\leq / \geq$ survive the passage to sup/inf.

Beyond the hints. Step 0 (existence of $\limsup$, $\liminf$ as genuine limits, via the Monotone Convergence Theorem) is entirely absent from the hints sheet; we supply it before either direction. We also carry out the full sandwich inequalities of both directions explicitly, rather than leaving them as “compare” and “bound.”

$\{a_n\} \text{ converges } \iff \liminf a_n = \limsup a_n,$ $\text{and then } \liminf a_n = \limsup a_n = \lim a_n.$

Question A2 (The Monotone Convergence Theorem via the Divergence Criterion)

Problem

Another way to prove the Monotone Convergence Theorem is via the Divergence Criterion.

- Prove that if $\{a_{n_k}\}$ and $\{a_{m_k}\}$ are two convergent subsequences of a monotone sequence $\{a_n\}$, then $\lim_{k \rightarrow \infty} a_{n_k} \geq \lim_{k \rightarrow \infty} a_{m_k}$.
- Hence use the Divergence Criterion to deduce that if, in addition, $\{a_n\}$ is bounded, then it is convergent.

(Tutorial 3, Additional Question 2; Lectures 7 and 8. Do not use the Monotone Convergence Theorem.)

Solution

Method 1: Leapfrogging one subsequence past the other

Official hint. (a) Use monotonicity to find a subsequence of $\{a_{n_k}\}$ whose terms are bounded below by corresponding terms of $\{a_{m_k}\}$; then apply Subsequential Limits. (b) Show that any two convergent subsequences of $\{a_n\}$ must have the same limit.

Idea. (Scratch work – not part of the written answer.) WLOG $\{a_n\}$ is increasing. Since $n_i \rightarrow \infty$, for each k we can find (later and later) indices $n_{i(k)} \geq m_k$; by monotonicity $a_{n_{i(k)}} \geq a_{m_k}$, and taking $k \rightarrow \infty$ with the Order Limit Theorem gives $\lim a_{n_{i(k)}} \geq \lim a_{m_k}$ directly. Relabelling the two given subsequences and re-running the same lemma gives the reverse inequality too, which is exactly why part (b) then forces equality.

Part (a).

Proof. Assume without loss of generality that $\{a_n\}$ is increasing (if $\{a_n\}$ is decreasing, apply the increasing case to $\{-a_n\}$ and negate). Write $A := \lim_{k \rightarrow \infty} a_{n_k}$ and $B := \lim_{k \rightarrow \infty} a_{m_k}$.

We construct a strictly increasing sequence of indices $i(1) < i(2) < i(3) < \dots$ in $\mathbb{N}$ such that $n_{i(k)} \geq m_k$ for every k . Since $\{n_i\}_{i=1}^\infty$ is strictly increasing, $n_i \geq i \rightarrow \infty$, so for each $k \in \mathbb{N}$ there exists $i \in \mathbb{N}$ with $n_i \geq m_k$; let $i(1) := \min\{i \in \mathbb{N} : n_i \geq m_1\}$. Having chosen $i(k)$, since $n_i \rightarrow \infty$ there again exists $i > i(k)$ with $n_i \geq m_{k+1}$; let $i(k+1) := \min\{i > i(k) : n_i \geq m_{k+1}\}$. By construction $i(1) < i(2) < i(3) < \dots$ and $n_{i(k)} \geq m_k$ for every $k \in \mathbb{N}$.

Since $\{i(k)\}_k$ is strictly increasing, $\{a_{n_{i(k)}}\}_k$ is a subsequence of $\{a_{n_k}\}_k$; by the Theorem on Subsequential Limits, $a_{n_{i(k)}} \rightarrow A$. Since $\{a_n\}$ is increasing and $n_{i(k)} \geq m_k$, monotonicity gives $a_{n_{i(k)}} \geq a_{m_k}$ for every $k \in \mathbb{N}$. Both sides converge ($a_{n_{i(k)}} \rightarrow A$, $a_{m_k} \rightarrow B$), so by the Order Limit Theorem, $A \geq B$; that is,

$$\lim_{k \rightarrow \infty} a_{n_k} \geq \lim_{k \rightarrow \infty} a_{m_k}. \square$$

Beyond the hints. The result of part (a), applied literally, holds for *either* labelling of the two given subsequences: relabelling $\{a_{m_k}\}$ as the “first” subsequence and $\{a_{n_k}\}$ as the “second” and re-running the identical construction gives $B \geq A$ as well. Combined with $A \geq B$ from above, this forces $A = B$. This is the symmetry the hints sheet’s bare “ $\geq$ ” does not make explicit, and it is exactly the fact part (b) needs: *any two convergent subsequences of a monotone sequence share the same limit.*

Part (b).

Proof. Suppose, in addition to being monotone, $\{a_n\}$ is bounded. Suppose, for contradiction, that $\{a_n\}$ is divergent. By the Divergence Criterion (applicable since $\{a_n\}$ is bounded), $\{a_n\}$ has two subsequences $\{a_{n_k}\}$ and $\{a_{m_k}\}$ converging to different limits, say $A \neq B$. But by part (a), applied to this pair, $A \geq B$; applied with the labels swapped, $B \geq A$. Hence $A = B$, contradicting $A \neq B$.

Therefore $\{a_n\}$ is not divergent. Since every sequence is either convergent or divergent, $\{a_n\}$ is convergent. $\square$

Common slip. Citing the Monotone Convergence Theorem itself at any point – the whole point of A2 is an independent proof of it, so only the Divergence Criterion, Subsequential Limits, the Order Limit Theorem, and (inside the Divergence Criterion’s own, already-established proof) Bolzano-Weierstrass may be used. A second slip is stopping at part (a)’s bare “ $\geq$ ” without noting the labelling symmetry, which silently leaves part (b) without its needed “equal limits” fact.

Beyond the hints. We give the explicit leapfrogging index construction $i(k)$ for part (a) (the hint only says “find a subsequence... bounded below,” without the recursive construction), the symmetry argument that upgrades “ $\geq$ ” to “=,” and the direct contradiction write-up for part (b).

- (a) $\lim a_{n_k} \geq \lim a_{m_k}$, with equality by the labelling symmetry.
 (b) monotone + bounded $\implies$ convergent.

Question A3 (The Bolzano-Weierstrass Theorem via Peak Terms)

Problem

Another way to prove the Bolzano-Weierstrass Theorem is via the Monotone Convergence Theorem. A useful notion for this is a *peak term*: a term a_m of a sequence $\{a_n\}$ is called a peak term if it is an upper bound for $\{a_n : n \in \mathbb{N} \text{ and } n \geq m\}$.

- Show that if a sequence $\{a_n\}$ has infinitely many peak terms, then the peak terms form a decreasing subsequence.
- Show that if a sequence $\{a_n\}$ does not have infinitely many peak terms, then it has an increasing subsequence.

The above show that every sequence $\{a_n\}$ contains a monotone subsequence. Use this, and the Monotone Convergence Theorem, to deduce the Bolzano-Weierstrass Theorem.

[Remark: The above two questions, together with the proof of the Divergence Criterion via the Bolzano-Weierstrass Theorem, show that the Divergence Criterion, the Bolzano-Weierstrass Theorem, and the Monotone Convergence Theorem are equivalent to each other for ordered fields.] (*Tutorial 3, Additional Question 3; Lectures 8 and 9.*

Do not use the Bolzano-Weierstrass Theorem.)

Solution

Method 1: Peak terms, then the Monotone Convergence Theorem

Official hint. (a) Use the definition of peak terms to show they form a decreasing subsequence. (b) Finitely many peak terms means some tail has no peak term; use the negation of the definition on this tail to recursively form an increasing subsequence.

Idea. (*Scratch work – not part of the written answer.*) a_m is a peak term means $a_m \geq a_n$ for every $n \geq m$. If two peak terms a_{m_1}, a_{m_2} have $m_1 < m_2$, then $a_{m_1} \geq a_{m_2}$ automatically, since $m_2 \geq m_1$ is one of the indices a_{m_1} dominates. If only finitely many peak terms exist, every term beyond the largest peak-term index is eventually overtaken, which lets us keep picking larger and larger values.

Part (a).

Proof. Suppose $\{a_n\}$ has infinitely many peak terms. List their indices in increasing order as $m_1 < m_2 < m_3 < \dots$ (possible since an infinite subset of $\mathbb{N}$ can be enumerated in increasing order); this is a strictly increasing sequence of natural numbers, so $\{a_{m_j}\}_{j=1}^{\infty}$ is a subsequence of $\{a_n\}$.

Fix $j \in \mathbb{N}$. Since m_j is a peak term, a_{m_j} is an upper bound for $\{a_n : n \geq m_j\}$; that is, $a_{m_j} \geq a_n$ for all $n \geq m_j$. In particular, since $m_{j+1} > m_j$, we get $a_{m_j} \geq a_{m_{j+1}}$. As $j \in \mathbb{N}$ was arbitrary, $a_{m_1} \geq a_{m_2} \geq a_{m_3} \geq \dots$, so $\{a_{m_j}\}$ is a decreasing subsequence of $\{a_n\}$. $\square$

Part (b).

Proof. Suppose $\{a_n\}$ does not have infinitely many peak terms; that is, the set of peak-term indices is finite (possibly empty). Let M_0 denote the largest peak-term index if at least one peak term exists, and $M_0 := 0$ otherwise; in either case, every index $m > M_0$ is *not* a peak-term index.

By the negation of the definition of a peak term: for $m > M_0$, a_m is not an upper bound for $\{a_n : n \geq m\}$, so there exists $n > m$ with $a_n > a_m$ (an index $n = m$ trivially fails to violate $a_m \geq a_n$, so the violating index must satisfy $n > m$).

We construct a strictly increasing index sequence $n_1 < n_2 < n_3 < \dots$ recursively. Let $n_1 := M_0 + 1$; since $n_1 > M_0$, a_{n_1} is not a peak term. Having chosen $n_k > M_0$ with a_{n_k} not a peak term, the above gives some $n_{k+1} > n_k$ with $a_{n_{k+1}} > a_{n_k}$; since $n_{k+1} > n_k > M_0$, $a_{n_{k+1}}$ is again not a peak term, so the construction continues. By induction this defines n_k for every $k \in \mathbb{N}$, with $n_1 < n_2 < n_3 < \dots$ (so $\{a_{n_k}\}$ is a subsequence of $\{a_n\}$) and $a_{n_1} < a_{n_2} < a_{n_3} < \dots$, i.e. a strictly increasing subsequence. $\square$

Step: every sequence has a monotone subsequence. Given any sequence $\{a_n\}$, either it has infinitely many peak terms, in which case part (a) exhibits a decreasing subsequence, or it does not, in which case part (b) exhibits an increasing subsequence. Either way, $\{a_n\}$ has a monotone subsequence.

Deducing the Bolzano-Weierstrass Theorem.

Proof. Let $\{a_n\}$ be a bounded sequence, say $|a_n| \leq M$ for all $n \in \mathbb{N}$. By the Step above, $\{a_n\}$ has a monotone subsequence $\{a_{n_k}\}$. Since $\{a_{n_k} : k \in \mathbb{N}\} \subseteq \{a_n : n \in \mathbb{N}\}$, we also have $|a_{n_k}| \leq M$ for every k , so $\{a_{n_k}\}$ is bounded. Being both monotone and bounded, the Monotone Convergence Theorem gives that $\{a_{n_k}\}$ converges. Therefore, this exhibits a convergent subsequence of $\{a_n\}$, which is exactly the statement of the Bolzano-Weierstrass Theorem. $\square$

Worked example. Let $a_n := 1/n$, decreasing. Every index m is a peak term here: $a_m = 1/m \geq 1/n = a_n$ for all $n \geq m$, since $1/n$ is decreasing. So there are infinitely many peak terms (every index), and part (a) applies directly: the peak-term subsequence is $\{a_m\}$ itself, already decreasing.

Now let $a_n := n$, increasing. No index m is a peak term, since $a_m = m < m+1 = a_{m+1}$ shows a_m is not an upper bound for the tail. So $M_0 = 0$ and part (b)'s construction gives $n_1 = 1, n_2 = 2, n_3 = 3, \dots$ – the whole sequence is already the increasing subsequence, consistent with $\{a_n\}$ being monotone increasing throughout.

A more instructive case: $a_n := (-1)^n(1 + 1/n)$, i.e. $a_1 = -2, a_2 = \frac{3}{2}, a_3 = -\frac{4}{3}, a_4 = \frac{5}{4}, \dots$, bounded by 2. The even-indexed terms $1 + 1/n \downarrow 1$ are all peak terms restricted among themselves, but a term a_{2j} is not an upper bound for the whole tail past it, since later odd terms are negative and later even terms $1 + 1/(2j')$ for $j' > j$ are smaller than $a_{2j} = 1 + 1/(2j)$ – so in fact every even-indexed a_{2j} is a peak term (it exceeds all later terms, both the smaller positive evens and the negative odds), giving infinitely many peak terms and the decreasing subsequence $\{a_{2j}\} = \{1 + 1/(2j)\} \downarrow 1$, matching part (a); the Bolzano-Weierstrass Theorem then extracts this same convergent (in fact already convergent) monotone piece.

Common slip. Forgetting the boundedness hypothesis of the Bolzano-Weierstrass Theorem itself: the monotone subsequence extracted from peak terms is bounded *only because* the original sequence was assumed bounded; without that hypothesis the Monotone Convergence Theorem does not apply. Another slip in part (b) is picking n_{k+1} to merely satisfy $a_{n_{k+1}} > a_{n_k}$ without also verifying $a_{n_{k+1}}$ is itself not a peak term (needed to con-

tinue the recursion) – this follows automatically once $n_{k+1} > M_0$, which must be checked, not assumed.

Beyond the hints. The hints sheet does not verify that the peak-term indices, listed in increasing order, form a legitimate infinite strictly-increasing sequence (we note this follows from enumerating an infinite subset of $\mathbb{N}$); it only sketches part (b)’s recursion (“recursively form an increasing subsequence”) without executing it, and does not spell out the final deduction of Bolzano-Weierstrass – in particular, that the extracted monotone subsequence inherits boundedness from the original sequence before the Monotone Convergence Theorem can be applied. We supply all three.

Every sequence has a monotone subsequence;
bounded + monotone $\xRightarrow{\text{MCT}}$ convergent, proving Bolzano-Weierstrass.

Question A4 (The Monotone Convergence Theorem via the Axiom of Completeness)

Problem

Yet another way to prove the Monotone Convergence Theorem is directly from the Axiom of Completeness. To this end, relate a sequence $\{a_n\}$ with the set $A := \{a_n : n \in \mathbb{N}\}$. Prove that if $\{a_n\}$ is increasing and bounded, then the supremum a of the set A is the limit of $\{a_n\}$. (*Tutorial 3, Additional Question 4; Lectures 2 and 9. Use only the Axiom of Completeness (and results proved directly from it); do not use the Bolzano-Weierstrass Theorem or the Monotone Convergence Theorem.*)

Solution

Method 1: The ε -Criterion for Suprema

Official hint. (a) Use the ε -Criterion for Suprema. (b) Follow the proof of the Monotone Convergence Lemma.

Idea. (*Scratch work – not part of the written answer.*) The ε -Criterion for Suprema (Lecture 2, deck p. 24) says: $s = \sup A$ iff s is an upper bound for A and $\forall \varepsilon > 0 \exists x \in A$ $s - \varepsilon < x$. Apply it once to locate an index N with a_N close to a ; then monotonicity carries this closeness to every $n \geq N$, since increasing terms can only get closer to (but never exceed) the supremum.

Setup. Let $\{a_n\}$ be increasing and bounded, and let $A := \{a_n : n \in \mathbb{N}\}$. Since $\{a_n\}$ is bounded, A is nonempty and bounded above, so by the Axiom of Completeness, $a := \sup A$ exists.

Proof. We recall the ε -Criterion for Suprema: for $A \subseteq \mathbb{R}$ and $s \in \mathbb{R}$, $s = \sup A$ if and only if s is an upper bound for A and, for every $\varepsilon > 0$, there exists $x \in A$ with $s - \varepsilon < x$.

Let $\varepsilon > 0$ be given. Since $a = \sup A$, the ε -Criterion gives $x \in A$ with $a - \varepsilon < x$. Since $x \in A = \{a_n : n \in \mathbb{N}\}$, there is $N \in \mathbb{N}$ with $x = a_N$; so $a - \varepsilon < a_N$.

Let $n \in \mathbb{N}$ with $n \geq N$. Since $\{a_n\}$ is increasing, $n \geq N$ gives $a_n \geq a_N$, so, combined with $a - \varepsilon < a_N$,

$$a - \varepsilon < a_N \leq a_n.$$

Also, since $a = \sup A$ is (by definition) an upper bound for A and $a_n \in A$, we have $a_n \leq a < a + \varepsilon$. Combining,

$$a - \varepsilon < a_n < a + \varepsilon, \quad \text{i.e.} \quad |a_n - a| < \varepsilon,$$

for every $n \geq N$. Since $\varepsilon > 0$ was arbitrary, this is exactly the definition of $a = \lim_{n \rightarrow \infty} a_n$. Hence $\{a_n\}$ converges, and its limit is $a = \sup A$. $\square$

Worked example. Let $a_n := 2 - 1/n$, increasing ($a_1 = 1, a_2 = \frac{3}{2}, a_3 = \frac{5}{3}, \dots$) and bounded above by 2. Here $A = \{2 - 1/n : n \in \mathbb{N}\}$, and $\sup A = 2$: it is an upper bound (since $1/n > 0$), and for any $\varepsilon > 0$ the Archimedean Property gives N with $1/N < \varepsilon$, so $a_N = 2 - 1/N > 2 - \varepsilon$, matching the ε -Criterion. The proof above then shows directly that $a_n \rightarrow 2$, without ever invoking Bolzano-Weierstrass or the Monotone Convergence Theorem's usual two-step (BW + Monotone Convergence Lemma) proof.

Remark (tying A2, A3, A4 together). This tutorial gives three logically independent routes to the Monotone Convergence Theorem: A2 derives it from the Divergence Criterion (itself proved from Bolzano-Weierstrass in Lecture 8); A3 runs the implication the other way, deriving Bolzano-Weierstrass from the Monotone Convergence Theorem via peak terms; and A4 derives the Monotone Convergence Theorem directly from the Axiom of Completeness, bypassing Bolzano-Weierstrass entirely. Together with Lecture 8’s own proof of the Divergence Criterion from Bolzano-Weierstrass, this is exactly the cycle of implications the question’s remark (Additional Question 3) promises: the Divergence Criterion, the Bolzano-Weierstrass Theorem, and the Monotone Convergence Theorem are all derivable from one another (and, via A4, directly from the Axiom of Completeness), so any one of them may be taken as the starting point for an ordered field satisfying completeness.

Common slip. Forgetting to justify $a_n \leq a$ separately from $a - \varepsilon < a_n$ – both halves of $|a_n - a| < \varepsilon$ come from two *different* facts about $a = \sup A$ (being an upper bound, versus the ε -Criterion’s “not-quite-an-upper-bound-below” clause), not from one inequality alone. A second slip is trying to run this argument without monotonicity: the step “ $n \geq N \implies a_n \geq a_N$ ” is exactly where the increasing hypothesis is used, and without it the single index N found by the ε -Criterion would not control the entire tail.

Beyond the hints. The hints sheet names the ε -Criterion for Suprema and says “follow the proof of the Monotone Convergence Lemma” but restates neither; we give the ε -Criterion’s statement in full, and the complete proof it licenses – in particular, how the criterion’s “ $s - \varepsilon$ is not an upper bound” clause produces the index N with $a_N > a - \varepsilon$, and how monotonicity then extends this bound to every $n \geq N$, which is the entire content of the proof and is exactly what the hints sheet omits.

$$\boxed{\{a_n\} \text{ increasing and bounded} \implies \{a_n\} \rightarrow \sup\{a_n : n \in \mathbb{N}\}.}$$

Method Map

Q.	Core topic	Methods included
1	T/F: subsequences and convergence	Deleting-one-term direct proof; two-overlapping-subsequences route; counterexample construction; contrapositive of Subsequential Limits.
2	T/F: existence of prescribed subsequences	Explicit interleaved construction; Bolzano-Weierstrass on a bounded subsequence; monotone bounded/unbounded case split.
3	Shuffled sequence, Divergence Criterion	Boundedness + odd/even pigeonhole + Subsequential Limits + Uniqueness of Limits, assembled via the (bounded) Divergence Criterion.
4	Three overlapping subsequences	Common-index construction ($6k, 6k+3$); adaptation of Question 3's pigeonhole argument to evens/odds.
5	Constructing a null subsequence	Recursive index construction with $\varepsilon_k = 1/k$; direct ε - K convergence proof.
A1	Limsup/liminf iff convergence	Monotone Convergence Theorem (existence of $\limsup, \liminf$); Squeeze Theorem; Order Limit Theorem via sup/inf definitions.
A2	MCT via Divergence Criterion	Leapfrogging index construction; Order Limit Theorem; Divergence Criterion by contradiction.
A3	Bolzano-Weierstrass via peak terms	Peak-term case split (infinite/finite); recursive increasing-subsequence construction; Monotone Convergence Theorem.
A4	MCT via Axiom of Completeness	ε -Criterion for Suprema; monotonicity extends a single close index to the whole tail.

Technique Summary

- Deleting or reindexing finitely many terms never changes convergence.** A tail of a sequence converges to the same limit as the whole sequence, and conversely (Q1(a) Method 1).
- Two subsequences sharing a common further subsequence must share a limit.** Apply the Theorem on Subsequential Limits twice (once to each), then Uniqueness of Limits (Q1(a) Method 2, Q3, Q4).
- The Divergence Criterion needs boundedness first.** Establish it explicitly before invoking the (bounded) Criterion by contrapositive (Q3, Q4, A2).
- “Infinitely many” is not exhausted by finitely many earlier choices.** The recursive-construction workhorse: remove a finite initial segment from an infinite set and it stays infinite (Q5, A3(b)).
- Monotone sequences: any two convergent subsequences agree.** Proved two ways in this tutorial – via MCT + Subsequential Limits in the bounded case (Q2(c)), and via a leapfrogging index construction + the Order Limit Theorem, with no monotone-convergence assumption (A2).

6. **The Divergence Criterion, Bolzano-Weierstrass, and the Monotone Convergence Theorem are logically equivalent** for ordered fields once the Axiom of Completeness is assumed: A2 gets MCT from the Divergence Criterion, A3 gets Bolzano-Weierstrass from MCT, A4 gets MCT directly from the Axiom of Completeness, and Lecture 8 already gets the Divergence Criterion from Bolzano-Weierstrass.

Final Answers

1. (a) True. (b) False. (c) True.
2. (a) True. (b) False. (c) False.
3. $\{c_n\}$ converges, to $L = \lim a_n = \lim b_n$.
4. $\{a_n\}$ converges, to $L = \lim a_{2k} = \lim a_{2k+1} = \lim a_{3k}$.
5. Such a subsequence exists; constructed via $\varepsilon_k = 1/k$, converging to 0.
6. $\{a_n\}$ converges $\iff \liminf a_n = \limsup a_n$, and then both equal $\lim a_n$.
7. (a) $\lim a_{n_k} \geq \lim a_{m_k}$ (equality by symmetry). (b) monotone + bounded $\implies$ convergent.
8. Every sequence has a monotone subsequence; bounded + monotone $\implies$ convergent (MCT), proving Bolzano-Weierstrass.
9. $\{a_n\}$ increasing and bounded $\implies \{a_n\} \rightarrow \sup\{a_n : n \in \mathbb{N}\}$.